Siddhant Kumar Upmanyu

Senior Systems Architect · Platform & SRE

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SUMMARY

The only infrastructure engineer at a smart-home IoT company for 4+ years. Own the cloud side end to end (architecture, reliability, cost, security and data) for a platform serving 10,000+ hubs and about 1 million connected devices. Write the protocol and API specs that a core team of about 10 firmware, mobile and backend engineers build against. Took over a manually deployed legacy stack, made it stable with automated, zero-downtime releases, and cut the monthly cloud bill by about 40%.

SKILLS

Cloud & Infrastructure: Google Cloud Platform (GCP: Compute Engine, Cloud Run, Cloud SQL, Cloud Storage, IAM, VPC), AWS (EKS), Kubernetes (k8s), Helm, Docker, Terraform, OpenTofu, Infrastructure as Code (IaC), Proxmox, cloud-init

Reliability & Delivery: CI/CD, GitHub Actions, GitOps, blue-green and canary releases, feature flags, automated rollback, incident response, root cause analysis (RCA), high availability, disaster recovery, cost optimization (FinOps)

Observability & Performance: Grafana, Loki, Prometheus, VictoriaMetrics, VictoriaLogs, Grafana Alloy, OpenTelemetry, distributed tracing, low-latency and JVM tuning

Linux, Networking & Security: Linux (Debian, Ubuntu), systemd, L3/L4 networking, TCP/UDP, Nginx, DNS, iptables, nftables, NAT64, rate limiting, DDoS mitigation, TLS, mTLS, PKI, SSO, zero trust

Data (OLTP & OLAP): PostgreSQL, MongoDB, Redis (Pub/Sub), Valkey, ClickHouse, high-ingestion pipelines, query optimization, indexing, data migration

Languages & Design: Go, Kotlin, Java, Python, Rust, Bash, SQL, C, Spring Boot, gRPC, Protocol Buffers, REST APIs, microservices, system design, API and protocol design, TDD

EXPERIENCE

eGlu Smart Homes (WiZNSystems)

Bengaluru, India

Senior Systems Architect (Platform & Infrastructure) — **Apr 2026 – Present**
Senior Software Engineer — **Apr 2024 – Mar 2026**
Software Engineer — **Mar 2022 – Mar 2024**

- Write the protocol and API specs that firmware, mobile and backend teams build against before each release, and lead R&D on the next-generation hub-to-cloud protocol.
- Designed the backend hiring process from scratch and hired a backend engineer.
- Took over a manually deployed legacy stack in 2023: replaced manual SFTP uploads with automated CI/CD and upgraded Spring 4 on Java 8 to Spring Boot 2.7 on Java 17.
- Made zero-downtime releases the default: blue-green cutovers using iptables, deploy scripts with health checks and automatic rollback, a staging environment that mirrors production, and feature flags for gradual rollouts.
- Fixed the root causes of recurring outages: file-descriptor limits that blocked new threads and SSH logins, a service leaking MongoDB connections, and stale Redis connections that silently dropped messages.
- Cut the monthly cloud bill by about 40% from its peak, with the new ClickHouse server still running, through committed use discounts, storage tuning and moving images out of the database.
- Split the marketing website out of the shared production Google Cloud project into its own project, isolating its access, billing and blast radius from the device platform. Moved it off a VM to Cloud Run, Cloud SQL and Cloud Storage, fully managed with Terraform, and served static assets straight from the bucket so Cloud Run only handles dynamic requests, cutting its runtime and cost.
- Traced disk I/O thrashing on production servers to raw telemetry log dumps. Rescheduling backups only shifted the spikes, so moved telemetry into ClickHouse with a Go ingestion service (gRPC), ending the I/O contention and cutting storage by over 95% (largest dataset: 90 GB to 3.7 GB).
- Moved image data out of MongoDB into Cloud Storage with zero downtime, shrinking the database 54x (3.6 GB to 67 MB) and its memory use by 89% (4.6 GB to 500 MB). Removed redundant indexes and added targeted ones to stop full collection scans.
- Connected IPv6-only cellular hubs to IPv4 backend services with in-kernel NAT64 (Jool and nftables), without adding a proxy. Added kernel-level rate limiting to drop abusive traffic before it reaches the application.
- Removed the need for SSH access to production: first built a Go tool for reading logs over mutual TLS with a private certificate authority, then moved to centralized logging and monitoring (Grafana, Loki, VictoriaMetrics) behind single sign-on (SSO).
- Maintain the custom UDP protocol that hubs use to talk to the cloud, and the service behind it, built for peaks of up to 50,000 packets per second. Tuned it for low latency, from kernel packet handling to an adaptive JVM thread pool and allocation-free decoding.

PROJECTS

gke-lab — Kubernetes migration prototype on GKE github.com/sku0x20/gke-lab

In progress. Prototyping the IoT platform's move from VMs to Kubernetes: a private GKE Autopilot cluster built with OpenTofu, running a 3-member MongoDB replica set (Percona operator) and Valkey, with secrets from Secret Manager via Workload Identity. Includes FinOps cost notes and a staged plan to self-manage nodes and the control plane.

avoid — Minimal Linux distribution github.com/sku0x20/avoid

Bootable Linux distribution based on Void Linux, built for server disaster recovery and small headless machines.

stopgap — Kotlin microservice framework github.com/sku0x20/stopgap

Built on Helidon SE and Java virtual threads. Published on Maven Central, with unit, integration and Docker end-to-end testing.

assertgo — Assertion library for Go github.com/sku0x20/assertgo

Type-safe, fluent assertions for Go tests using generics, with negation, custom comparators and custom matchers. Released v1.1.0 with CI.

PUBLICATIONS

Connection-Agnostic Presence Tracking for Stateless Distributed Backends. Zenodo preprint. doi.org/10.5281/zenodo.21717242

EDUCATION

Bachelor of Computer Applications (BCA) 2019 – 2022